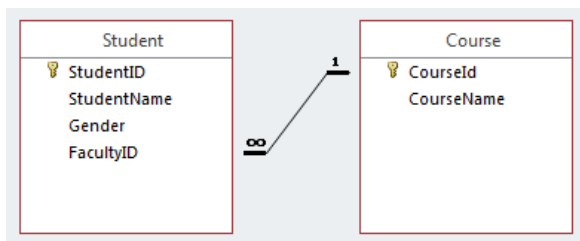


INF10025 Data Management and Analytics

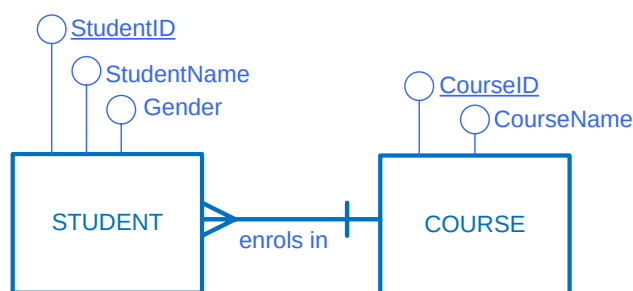
Tutorial 5 – Discussion Questions for Week 5

1. How are Access Relationship Diagrams different from Entity Relationship Diagrams?
Use these two diagrams as a reference if you wish.

ARD:



ERD:



2. How do Natural Keys and Surrogate Keys differ?
3. Read the following text from: <http://stackoverflow.com/questions/63090/surrogate-vs-natural-business-keys>

I hate surrogate keys in general. They should only be used when there is no quality natural key available. It is rather absurd when you think about it, to think that adding meaningless data to your table could make things better. Here are my reasons:

1. When using natural keys, tables are clustered in the way that they are most often searched thus making queries faster.
2. When using surrogate keys you must add unique indexes on logical key columns. You still need to prevent logical duplicate data. For example, you can't allow two Organizations with the same name in your Organization table even though the pk is a surrogate id column.
3. When surrogate keys are used as the primary key it is much less clear what the natural primary keys are. When developing you want to know what set of columns make the table unique.
4. In one to many relationship chains, the logical key chains. So for example, Organizations have many Accounts and Accounts have many Invoices. So the logical-key of Organization is OrgName. The logical-key of Accounts is OrgName, AccountID. The logical-key of Invoice is OrgName, AccountID, InvoiceNumber. When surrogate keys are used, the key chains are truncated by only having a foreign key to the immediate parent. For example, the Invoice table does not have an OrgName column. It only has a column for the AccountID. If you want to search for invoices for a given organization, then you will need to join the Organization, Account, and Invoice tables. If you use logical keys, then you could Query the Organization table directly.
5. Storing surrogate key values of lookup tables causes tables to be filled with meaningless integers. To view the data, complex views must be created that join to all of the lookup tables. A lookup table is meant to hold a set of acceptable values for a column. It should not be codified by storing an integer surrogate key instead. There is nothing in the normalization rules that suggest that you should store a surrogate integer instead of the value itself.
6. I have three different database books. Not one of them shows using surrogate keys.

Ken

Response 1: "I hate surrogate keys, except when they are necessary." They are necessary when the enterprise uses a natural key that is subject to a lot of errors, and are unwilling to tolerate a database that's impacted by those errors. – **Walter Mitty**

Response 2: I have written and maintained dozens of applications. The ones with the most data-related problems were those using natural keys. – **Falcon**

Response 3: #6 is actually a pretty compelling argument. Though I may be biased because I dislike surrogate keys. **James King**

Response 4: Adding to #3.: in other words, you instantly know which are the mandatory fields in a user form to be able to create another row in a table. – **DanMan**

Response 5: Some of your points assume the surrogate key has to be the PK or has to be the clustered column--not true. Your points 1 and 5 ignore the fact that integers are 4 bytes and natural keys are almost always many, many more bytes... which translates into much worse read performance, which makes queries slower, not faster. – **ErikE**

Response 6: Another reason against natural keys (examples: Atomic Numbers, VINs, etc), the business logic can change that increases the type of data. For example - Before: Tracking charges of Atoms, After: Tracking charges of Atoms and Compounds. Before: Tracking Motor Vehicles for load capacity. After: Adding planes, boats, bikes and people for load carrying capacity. – **forforf**

Response 7: I guess you don't have any tables where the primary key is composed even partially from 1) any attribute that can and will change), or 2) from user input (e.g. dynamically generated lookup lists). If you can't guarantee key immutability, then you will have to update all these entity relationships by code or by manual "fix" scripts. If you never had to do that... I guess your database is both surrogate key-less and unusual. – **code4life**

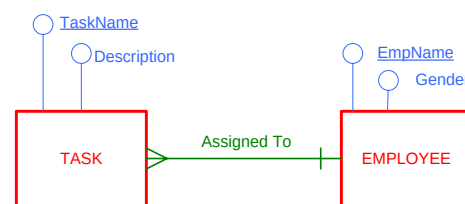
a. Is there are winner from this discussion?

b. What conclusions do you draw from the above discussion?

4. Why do different organisations in the same industry have different business rules?

5. Consider this ERD:

- Specify the relationship between Task and Employee by reading Left to Right.
- Specify the relationship between Task and Employee by reading Right to Left.
- When converted to a Relational Schema will any of the relations have a foreign key? Why or why not?



6. If you answered Yes to the above question, which relation will have the foreign key?

7. Convert the above ERD to a Relational Schema

8. Modify the above Relational Schema so that *surrogate keys are used* instead of natural keys

9. Write the SQL *Create Table* statements based on the above relational schema.

10. Write the SQL *Insert* statements to add this data to table

1	Fred Bloggs	Male
2	Sue Davis	Female
3	Emma Moore	Female

1	Cabling	Install Network Cable	2
2	Build PCs	Build 25 PCs for new office	3

11. For each of the following statements, indicate if the statement is valid or invalid

- a. **SELECT** Empname, Gender
- b. **SELECT** Employee
- c. **SELECT * FROM** Employee
- d. **SELECT** Fred **FROM** Employee
- e. **SELECT ALL FROM** Employee
- f. **SELECT** EmpName
FROM Employee
WHERE Gender Is M
- g. **SELECT** EmpName, Gender
ORDER BY EmpName DESC
- h. **SELECT * FROM** Employee
ORDER BY EmpName
- i. **SELECT** Empid, EmpName, Gender
WHERE Gender ='Female'
ORDER BY EmpId ASC

12. Consider the following SQL statement:

```
CREATE TABLE Person (  
  PerID      NUMBER      PRIMARY KEY,  
  PerName    VARCHAR(50)  
);
```

- a. How many tables would be created if you were to execute this statement in iSQL Jr?
- b. What occurs if you were to execute this statement a second or third time?

13. Consider this script of SQL statements:

```
DROP TABLE Worker;  
CREATE TABLE Worker (  
  WID        NUMBER      PRIMARY KEY,  
  WName      VARCHAR(50)  
);
```

- a. What is the significance of the ; (semi colon) symbol in this script?
- b. What happens whenever the first statement of the script is executed?
- c. What occurs if you were to execute this script a second or third time?